Ryan Wong

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Research Interests

Computer architecture; memory & storage systems; emerging memory technologies; hardware accelerators for machine learning, scientific computing/SciML, and database systems

Education

University of Illinois Urbana-Champaign — **Urbana, Illinois**
Ph.D. in Computer Science, GPA: 4.0/4.0 — *Expected 2027*
Advisor: Saugata Ghose

University of Rochester — **Rochester, New York**
M.S. in Electrical Engineering — *2020*
Advisor: Engin Ipek

University of Rochester — **Rochester, New York**
B.S. in Computer Science/B.A. in Chemistry — *2018*
Distinction in Chemistry

Awards & Distinctions

MICRO PhD Forum: TCuArch — 2026
ML and Systems Rising Star: MLCommons — 2026
Heidelberg Laureate Forum Young Researcher: Heidelberg Laureate Forum Foundation — 2026
Mavis Future Faculty Fellowship: University of Illinois — 2025–2026
Spot Award: Data Storage Architectures: Sandia National Laboratories — 2025
Outstanding Teaching Assistant: University of Illinois — 2022
Hopeman Fellowship: University of Rochester — 2019–2020

Publications & Peer-Reviewed Workshops

[IN PREPARATION] R. Wong, A. Das, D. Meeks, B. Feinberg, and S. Ghose, "Title Redacted for Blind Review", *60th International Symposium on Microarchitecture* **(MICRO)**, 2028.

A. Tyagi, M.S.Q. Truong, **R. Wong**, Y. Sun, D. Xiong, and S. Ghose, "Cross-Stack Design Considerations for Viable Processing-Using-Memory Architectures", *33rd International Symposium on High-Performance Computer Architecture* **(HPCA)**, 2027.

F. Seiler, H. Hu, **R. Wong**, K. K. Güclü, Y. Sun, S. Ghose, J. Aghassi-Hagmann, and N. TaheriNejad, "Cross-Layer Design Approaches of Stateful Digital Memristive In-Memory Computing", *invited paper in 39th International System-on-Chip Conference* (**SoCC**), 2026.

R. Wong, B. Feinberg, and S. Ghose, "DARTH-PUM: A Hybrid Processing-Using-Memory Architecture", *31st International Conference on Architectural Support for Programming Languages and Operating Systems* **(ASPLOS)**, 2026.
Extended Abstract in *2nd Cross-disciplinary Conference on Memory-Centric Computing* (CCMCC), 2026.

A. Das, **R. Wong**, A. Tyagi, and S. Ghose, "BRIE-RAM: Exploring Biodegradable Processing-Using-Memory for Sustainable Edge Sensing", *2nd Workshop on Ethical Systems and Architecture Design* **(HotEthics)** co-located with ASPLOS, 2026.

R. Wong, N. Kim, A. Das, K. Higgs, E. Ipek, S. Agarwal, S. Ghose, and B. Feinberg, "ANVIL: An In-Storage Accelerator for Name–Value Data Stores", *52nd International Symposium on Computer Architecture* **(ISCA)**, 2025.

R. Wong, N. Kim, K. Higgs, E. Ipek, S. Agarwal, S. Ghose, and B. Feinberg, "TCAM-SSD: A Framework for Search-Based Computing in Solid-State Drives", *15th Non-Volatile Memories Workshop* **(NVMW)**, 2024. *Extended paper available on arXiv:* `https://arxiv.org/abs/2403.06938`

B. Feinberg, **R. Wong**, T. P. Xiao, C. H. Bennett, J. N. Rohan, E. G. Boman, M. J. Marinella, S. Agarwal, and E. Ipek, "An Analog Preconditioner for Solving Linear Systems", *27th International Symposium on High-Performance Computer Architecture* **(HPCA)**, 2021.

B. Feinberg, B. Heyman, D. Mikhailenko, **R. Wong**, A. Ho, and E. Ipek, "Commutative Data Reordering: A New Technique to Reduce Data Movement Energy on Sparse Inference Workloads", *47th International Symposium on Computer Architecture* **(ISCA)**, 2020.

B. Feinberg, B. Heyman, D. Mikhailenko, **R. Wong**, and E. Ipek, "Reducing Data Movement Energy via Commutative Data Reordering", *Government Microcircuit Applications & Critical Technology Conference* **(GOMACTech)**, 2019.

Technical Reports

S. Agarwal, B. Feinberg, J. N. Rohan, T. P. Xiao, C. H. Bennett, E. G. Boman, M. J. Marinella, **R. Wong**, B. C. Heyman, D. Mikhailenko, A. C. Ho, and E. Ipek, "High Precision Sparse and Dense Analog Matrix Multiplication", *Sandia Report*, SAND2021-12424, 2021.

Invited Talks, Tutorials, and Workshops

Organizer for "Accelerating Your Job Search Workshop", Co-located with *MICRO 2026*.

Panelist for "Life in Grad School", Undergraduate Architecture Mentoring (uArch) Workshop, Co-located with *ISCA 2026*.

Invited Speaker for "Enabling Near-Accelerator Performance With Generic Processing-Using-Memory Architectures", *5th Workshop on Memory-Centric Computing Systems (MCCSys)*, Co-located with *ASPLOS 2026*.

Organizer for "Simulation for Processing-Using-Memory Systems (PUMPS)", Co-located with *ISCA 2024*.

Panelist for "Life in Grad School", Undergraduate Architecture Mentoring (uArch) Workshop, Co-located with *ISCA 2024*.

Professional Experience

Graduate R&D Intern — **Sandia National Laboratories**
Scalable Computer Architectures — *2024–2025*
Mentor: Ben Feinberg

Radiation Hardened CMOS — *2019–2021*
Co-advisors: Ben Feinberg, Sapan Agarwal

Computer Systems Architecture Laboratory — **University of Rochester**
(Graduate) Research Assistant — *2017–2021*
Advisor: Engin Ipek

NSF-Research Experience for Undergraduates — **Salisbury University**
Research Assistant — *Summer 2018*
Advisor: Lei Zhang

ICODES Test Group — **Tapestry Solutions**
Software Tester — *Summer 2016, 2017*

Teaching

CS 233H: Computer Architecture Honors — **University of Illinois**
Instructors: Ryan Wong & Prof. Geoffrey Herman — *Fall 2023*
Overall teaching rating 4.63/5, Overall course rating 4.63/5 (16 responses)
On List of Teachers Ranked as Excellent by Their Students

CS 233H: Computer Architecture — **University of Illinois**
Instructors: Profs. Geoffrey Herman & Saugata Ghose — *Fall 2022*

Instructors: Profs. Geoffrey Herman & Saugata Ghose — *Fall 2021*
Outstanding TA Award
Overall teaching rating: 4.38/5 (8 responses)
On List of Teachers Ranked as Excellent by Their Students

ECE 201/401: Advanced Computer Architecture — **University of Rochester**
Instructor: Prof. Engin Ipek — *Fall 2019*

ECE 200/400: Computer Organization — **University of Rochester**
Instructor: Prof. Engin Ipek — *Spring 2019*

CSC 172: Data Structures and Algorithms — **University of Rochester**
Instructor: Prof. Tamal Biswas — *(Head Workshop Leader) Spring 2018*

Instructor: Prof. Ted Pawlicki — *Spring 2017*

CSC 242/442: Artificial Intelligence — **University of Rochester**
Instructor: Prof. George Ferguson — *Fall 2017*

CSC 171: Introduction to Computer Science — **University of Rochester**
Instructor: Prof. Ted Pawlicki — *(Head Workshop Leader) Fall 2017*

Instructor: Prof. George Ferguson — *Fall 2016*

Mentoring

Dylan Meeks — **University of Illinois**
Thermal Modeling for Processing-Using-Memory — *2026–Present*

Paige Nothelfer — **University of Illinois**
Thermal Modeling for Processing-Using-Memory — *2025–Present*

Aniket Das — **University of Illinois**
Hybrid Analog–Digital Processing-Using-Memory — *2025–Present*
Senior Thesis: Reliability for In-Flash Processing-Using-Memory — *2024–2025*

Nandini Rao — **University of Illinois**
Software Defined Caches — *2025–2026*

Abhinil Dutt — **University of Illinois**
Senior Thesis: Software Defined Caches — *2023–2025*

Jenny Liang — **University of Illinois**
Adaptive Cache Hierarchies — *2023–2025*

Rahul Prabhu — **University of Illinois**
Senior Thesis: PUM Architectures — *2023–2024*

Jiwon (Julie) Lee — **University of Illinois**
Senior Thesis: Adaptive Cache Hierarchies — *2022–2023*

Kevin Higgs — **University of Illinois**
ISUR: In-Storage Computing — *2022–2023*

Nikita Kim — **University of Rochester**
In-Storage Computing — *2019–2022*

Service

- **Computer Architecture Student Association (CASA) Steering Committee Member**: 2022–Present
- **Computer Architecture Student Association (CASA) Co-Chair**: 2026–Present
- **Meet-A-Senior-Architect/Meet-A-Senior-Student (MaSA/MaSS) Chair**: ISCA 2026, MICRO 2026 (co-chair)
- **uArch Mentor**: ISCA 2024, ISCA 2025, ISCA 2026
- **CS Graduate Student Ambassador (Illinois)**: 2023, 2024, 2025, 2026
- **Illinois Scholars Undergraduate Research (ISUR) Mentor, DaRin Butz Mentor**: 2022, 2023, 2024
- **Program Committee/Reviewer**: MICRO 2026 (Student PC), CAL